FaceRankNet: An Interpretable Part-Based Graph Attention Network for Geometry-Driven Facial Beauty Prediction

Anonymous Author(s)
Department of Computer Science
Institution Name
City, Country
author@institution.edu

***Abstract*—Facial beauty prediction (FBP) models built on convolutional neural networks achieve strong accuracy but operate as opaque pixel-level regressors, leaving the question of *which* facial regions drive an aesthetic judgment to post-hoc saliency analysis. We present FaceRankNet, a facial beauty predictor that is interpretable by construction: rather than ingesting raw pixels, it operates on the geometry of 468 MediaPipe facial landmarks, partitions the face into five anatomical sub-graphs (left eye, right eye, nose, mouth, and jawline), and scores each region with a dedicated graph attention encoder. A learnable softmax over the five regions fuses the local scores into a global prediction, so the model emits a per-organ aesthetic contribution as a first-class output rather than an explanation reconstructed after the fact. Training combines a weak-supervision signal derived from a per-ethnicity *beauty-axis projection* of the landmark geometry with an anchor regression loss, a feature-level pairwise ranking loss, and a diversity-plus-boundary regularizer. On the SCUT-FBP5500 benchmark, FaceRankNet attains a test Pearson correlation of 0.6222, a mean absolute error of 0.4410, and a demographic parity difference of only 0.0855 between its Asian and Caucasian subgroups. The learned organ weights yield interpretable, non-degenerate per-region scores, and the geometry-only design structurally reduces the demographic disparity typical of pixel-based predictors without eliminating it. We analyze a systematic regression-to-the-mean failure mode at the rating extremes and use it, together with the ethical hazards of automated beauty scoring, to motivate a dedicated future-work and safety agenda.**

***Index Terms*—facial beauty prediction, graph attention networks, interpretability, weak supervision, algorithmic fairness, facial landmarks**

I. Introduction

Facial beauty prediction uses computer vision and machine learning to automate the assessment of human facial attractiveness from image data, with applications spanning cosmetic surgery planning, social and dating platforms, advertising, and the entertainment industry [1]. Public benchmarks such as SCUT-FBP5500 [2] and MEBeauty [3] collect attractiveness ratings from many annotators per image on a fixed scale, and these ratings are used to train models that emulate aggregate human judgment. Deep convolutional neural networks (CNNs) and, more recently, vision transformers have driven steady accuracy gains on these benchmarks [4]–[6].

Despite this progress, a central limitation persists: state-of-the-art FBP models are difficult to interpret. A pixel-level regressor can report *that* a face is attractive but cannot directly report *which* facial regions support that prediction. This interpretability gap matters for the very applications that make FBP attractive. In cosmetic surgery planning and personalized recommendation, the basis of an attractiveness judgment is the clinically relevant quantity; in any deployment, transparency is what allows practitioners to recognize bias, audit behavior, and improve the model [1], [7]. Recent work by Ibrahim, Ugail, and Ugail [1] confronts this gap directly, introducing a multi-method framework that combines permutation feature importance, a region-aggregated XRAI saliency technique they call Region Attribution, and individual-feature prediction to ask which regions a trained CNN relies on. Their analysis consistently identifies the eyes and nose as dominant, but it remains fundamentally *post-hoc*: the explanations are reconstructed from a model that was never built to expose them.

We take the complementary position that a beauty predictor should be interpretable by construction. FaceRankNet is built around three design commitments. First, the model never sees pixels; it operates on the geometry of 468 three-dimensional MediaPipe landmarks [8], which removes skin tone, lighting, and texture as direct inputs and is the structural mechanism by which we reduce demographic bias. Second, the face is explicitly partitioned into five anatomical organ sub-graphs, each scored by its own graph attention encoder [9], so that an aesthetic contribution per region is computed in the forward pass. Third, the global score is a learnable softmax-weighted sum of the five local scores, which keeps the output mathematically bounded to the rating scale and makes the fusion weights directly readable as the relative aesthetic importance of each organ.

The contributions of this paper are as follows. (i) We propose a geometry-only, part-based graph attention architecture for facial beauty prediction whose per-organ scores and fusion weights are interpretable by construction rather than by post-hoc attribution. (ii) We introduce a weak-supervision signal based on a per-ethnicity beauty-axis projection of the landmark

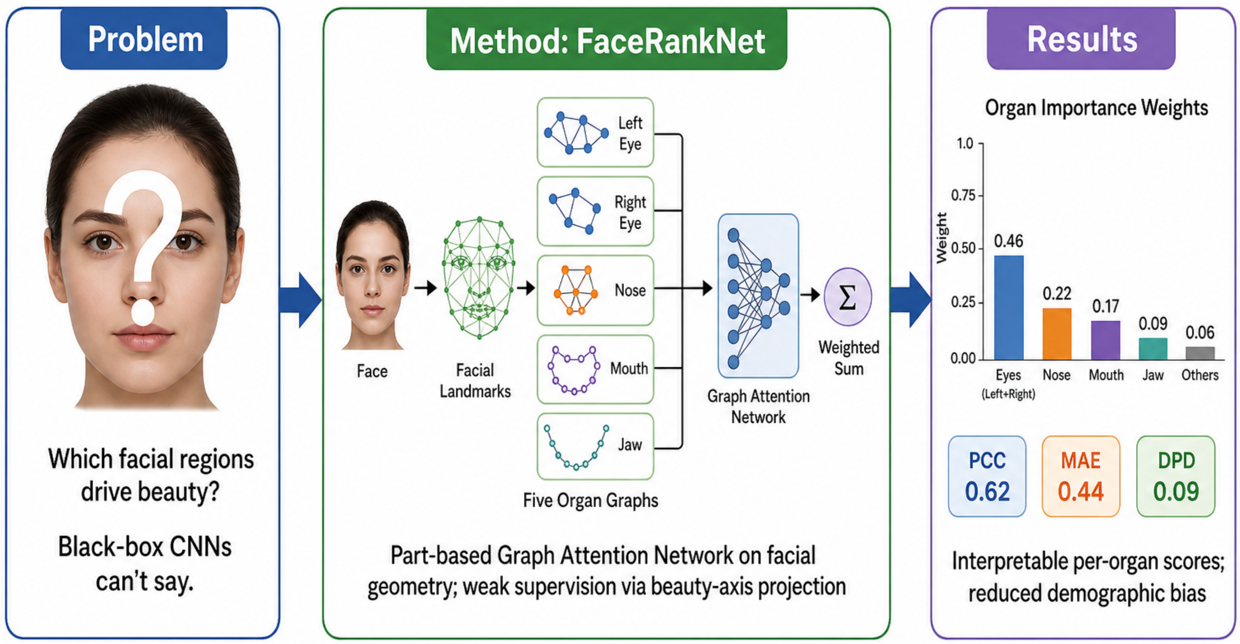

Fig. 1. Graphical abstract. FaceRankNet replaces a black-box pixel regressor with a part-based graph attention network defined on facial geometry. The five organ sub-graphs are scored independently and fused by a learnable softmax, producing both a global beauty score and an interpretable per-organ importance distribution while structurally reducing demographic bias.

geometry, and we show through an ablation that this directional formulation substantially outperforms distance-based and symmetry-based pseudo-labels. (iii) We report a complete evaluation on SCUT-FBP5500, including a per-ethnicity fairness breakdown and a per-bucket error analysis that exposes a regression-to-the-mean failure mode. (iv) Motivated by both that failure mode and the ethical hazards of automated beauty scoring, we articulate a future-work and safety agenda that the model's intrinsic interpretability is well placed to support.

II. Related Work

A. Geometric and Perceptual Accounts of Facial Beauty

Early attempts to quantify facial attractiveness emphasized geometric regularities such as bilateral symmetry and "averageness," the observation that faces close to the population mean tend to be rated as attractive [10], [11]. However, attractiveness is also strongly shaped by individual and cultural factors; aesthetic facial geometries differ across ethnicities, and there is no single universal standard of beauty [12], [13]. At the same time, despite this variation, raters on average reach substantial agreement on the attractiveness of a given face, which is what makes supervised FBP feasible at all. FaceRankNet inherits the geometric tradition by modeling landmark structure directly, but it operationalizes "averageness" as a learnable, per-ethnicity *direction* toward an attractive prototype rather than as a fixed symmetry or distance heuristic.

B. Deep Learning for Facial Beauty Prediction

Handcrafted descriptors and geometric ratios were the first machine-learning tools applied to FBP [14], [15], but deep CNNs quickly became the state of the art, taking a full-face image as input and regressing a beauty score [4], [5], [16]. Modeling the full rating distribution rather than only its mean has been shown to improve predictive performance [4]. A recent and especially relevant line of work fuses appearance with geometry: GeoFusion-Net [17] pairs an EfficientNet-B0 visual branch with a graph attention branch over the dataset's 86 two-dimensional landmarks, combining them through an attention-based fusion to reach a Pearson correlation of 0.9181 on SCUT-FBP5500. Such hybrids are accurate, but they retain a pixel branch and therefore consume photometric appearance directly, which makes them susceptible to the demographic confounds we seek to avoid. FaceRankNet departs from this line by discarding pixels entirely and reasoning only over three-dimensional landmark geometry with a part-based graph network, trading a degree of raw accuracy for intrinsic interpretability and reduced demographic dependence.

C. Interpretability for Vision Models and FBP

Interpretability is a mature subfield in machine learning. Permutation feature importance measures the performance drop when a feature is randomly shuffled [18]; Grad-CAM [19] and XRAI [20] produce spatial saliency maps for CNNs; and SHAP [21] and LIME [22] provide model-agnostic local attributions. Ibrahim *et al.* [1] adapt several of these to FBP, segmenting faces with landmarks and treating regions as features for permutation importance, aggregating XRAI saliency into region-level scores, and training per-region CNNs; across SCUT-FBP5500 and MEBeauty they find the eyes and nose to be the most influential regions and the chin the least. Our work is methodologically distinct: where these techniques explain an existing black box after training, FaceRankNet is an interpretable model in the sense advocated by Rudin [7], computing region contributions in the forward pass. This lets us compare our learned, intrinsic organ weights against their post-hoc regional rankings, providing a useful cross-check between the two paradigms.

D. Weak Supervision from Facial Geometry

Because dense per-region attractiveness labels do not exist, FaceRankNet supervises its organ encoders with geometric pseudo-labels. Rather than scoring a face by its distance from an average or prototype face, we project the landmark geometry onto a per-ethnicity *beauty axis* pointing from the population mean toward an attractive prototype. As Section IV shows empirically, this directional formulation aligns far better with human ratings than distance- or symmetry-based alternatives, echoing the perceptual finding that the *direction* of deviation toward an aesthetic ideal carries more signal than raw geometric distance from it.

III. Methodology

A. Dataset and Splits

We use SCUT-FBP5500 [2], which comprises 5,500 frontal facial images spanning two ethnicities (Asian and Caucasian) and two genders, each annotated with a holistic beauty score in $[1, 5]$ averaged over 60 raters. We derive the ethnicity attribute from the dataset filename convention. Following the benchmark protocol, we hold out a standard 80/20 train/test split (4,400 training and 1,100 test images). To enable principled model selection, we additionally carve a stratified 90/10 train/validation split from the training portion (stratified by rating bucket), yielding 3,960 training, 440 validation, and 1,100 test images. The model checkpoint with the best validation Pearson correlation is selected, and the test set is evaluated exactly once.

B. Preprocessing and Node Features

Each face is processed with the MediaPipe Face Mesh [8], which returns 468 three-dimensional landmark coordinates with consistent anatomical indexing. We normalize each landmark cloud by centering it on the centroid of all 468 points and dividing by the inter-ocular distance between the outer eye corners, removing translation and scale. Every node then carries a six-dimensional feature vector $(x, y, z, \Delta x, \Delta y, \Delta z)$, where the first triple is the normalized coordinate and the second triple is the per-node deviation from the Universal Average Face, defined as the mean landmark position over the training set. This deviation channel injects the averageness signal into every node, and the encoder's input projection is correspondingly $\mathrm{Linear}(6 \to 64)$.

We partition the landmarks into five organ sub-graphs: left eye, right eye, nose, mouth, and jawline, with the eyebrows merged into their respective eye sub-graphs. The two eyes are modeled as independent sub-graphs rather than a single combined region. Each sub-graph is made fully connected with self-loops so that attention can be computed over all intra-organ landmark pairs.

C. Weakly Supervised Beauty-Axis Pseudo-Labels

We generate per-organ pseudo-labels by projecting landmark geometry onto a learned aesthetic direction, computed separately for each ethnicity group e (hypothesis H1). For group e, let μ_e be the mean landmark configuration over its training faces and let π_e be the mean configuration of its top-30% rated training faces. The beauty axis is the difference vector $a_e = \pi_e - \mu_e$, pointing from the population average toward the attractive end of that group. For each training face f and organ o, we restrict a_e to the landmark indices of o and compute the scalar projection of $(\,\mathrm{coords}(f)_o - \mu_{e,o}\,)$ onto that organ-restricted axis. The per-organ projections are then percentile-ranked across the training set and remapped to the rating scale as $s_o^{\mathrm{psc}} = 1 + 4\,\mathrm{percentile\text{-}rank} \in [1, 5]$. Computing μ_e, π_e, and a_e per ethnicity is the entire mechanism of H1: Asian and Caucasian faces are each compared against their own group's prototype and axis. Before training proceeds, we apply a validation gate requiring a Spearman correlation $\rho > 0.2$ between the organ-averaged pseudo-score and the holistic ground-truth rating. Algorithm 1 summarizes the full procedure.

D. The FaceRankNet Architecture

Figure 2 summarizes the architecture. Each organ is encoded by an independent module, `OrganGAT`. The six-dimensional node features are projected to 64 dimensions and passed through a graph attention layer [9] with four attention heads, residual connections, and ELU activations; concatenating the heads yields 256-dimensional node embeddings. The node embeddings are pooled into a single organ embedding with gated attention pooling, implemented via the Deep Graph Library's `GlobalAttentionPooling` [23], [24] using a learnable gate $\mathrm{Linear}(256 \to 1)$. A read-out head, $\mathrm{Linear}(256 \to 32) \to \mathrm{ELU} \to \mathrm{Dropout}(0.1) \to \mathrm{Linear}(32 \to 1)$, produces a scalar that is squashed into the open rating interval as

$$\hat{y}_{\mathrm{organ}_i} = 4\,\sigma(x_i) + 1 \in (1, 5). \tag{1}$$

The five local scores are fused into a global prediction by a learnable softmax over per-organ logits w_i:

$$\hat{y}_{\mathrm{global}} = \sum_{i=1}^{5} \mathrm{softmax}(w_i)\,\hat{y}_{\mathrm{organ}_i}. \tag{2}$$

Algorithm 1 Per-ethnicity beauty-axis pseudo-labels

```
Require: training landmarks {c_f}, ratings {y_f}, ethnicity {e_f}, organs 𝒪
Ensure: per-organ pseudo-scores s^psc_{f,o} ∈ [1, 5]
for each ethnicity group e do
    μ_e ← mean{c_f : e_f = e} {population mean}
    π_e ← mean{c_f : e_f = e, y_f ∈ top 30%} {prototype}
    a_e ← π_e − μ_e {beauty axis}
end for
for each face f with ethnicity e = e_f and each organ o ∈ 𝒪 do
    p_{f,o} ← ⟨ c_{f,o} − μ_{e,o}, a_{e,o} ⟩ / ‖a_{e,o}‖
end for
for each organ o ∈ 𝒪 do
    r_{f,o} ← percentile-rank(p_{f,o}) over all faces
    s^psc_{f,o} ← 1 + 4 r_{f,o}
end for
s̄_f ← mean_o s^psc_{f,o}
if Spearman(s̄, y) ≤ 0.2 then
    abort {validation gate failed}
end if
return {s^psc_{f,o}}
```

Because each local score lies in $(1,5)$ and the fusion weights are non-negative and sum to one, the global score is mathematically guaranteed to lie in $[1,5]$. The architecture contains no cross-organ attention and no global multilayer perceptron; the fused score is a closed-form convex combination of interpretable parts, and the learned weights $\mathrm{softmax}(w_i)$ are read directly as each organ's relative aesthetic contribution.

E. Loss Function

The model is trained with a fixed-weight combination of three terms,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{reg}} + 1.0\,\mathcal{L}_{\text{rank}} + 0.01\,\mathcal{L}_{\text{div}}. \tag{3}$$

The weights are constants; we deliberately avoid adaptive schemes such as GradNorm [25], and an ablation with the ranking weight set to zero confirmed that the ranking term is required for organ-level fidelity. The anchor regression term is the mean squared error against the holistic rating,

$$\mathcal{L}_{\text{reg}} = \mathrm{MSE}(\hat{y}_{\text{global}}, y_{\text{gt}}). \tag{4}$$

The feature-level pairwise ranking term operates per organ and is a margin-masked logistic (RankNet-style [26]) loss,

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\text{kept}|}\sum_{o}\sum_{(A,B)} m_o(A,B)\,\log\!\big(1+e^{\hat{y}_o(B)-\hat{y}_o(A)}\big), \tag{5}$$

where the per-organ mask $m_o(A,B) = \mathbb{1}\big[\hat{y}_o^{\text{psc}}(A) - \hat{y}_o^{\text{psc}}(B) > \tau\big]$ retains only pairs whose beauty-axis pseudo-scores (Section III-C) differ by more than $\tau = 0.3$; the loss is implemented with $\log(1+e^{\cdot})$ for numerical stability.

TABLE I
TRAINING CONFIGURATION.

Setting	Value
Framework	PyTorch + DGL
Optimizer	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR schedule	none
Batch size	32
Epochs	50
Loss weights	λ_{reg}=1.0, λ_{rank}=1.0, λ_{div}=0.01
Fusion mode	softmax-weighted (Eq. 2)
Pair sampling	weighted (sqrt) + hard, rebuilt per epoch
Augmentation	landmark jitter σ=0.003
Model selection	best validation PCC

The diversity-plus-boundary term spreads the organ scores apart while preventing sigmoid saturation,

$$\mathcal{L}_{\text{div}} = -\mathrm{Var}_o(\hat{y}_o) + \mathrm{ReLU}(1.2 - \hat{y}_o) + \mathrm{ReLU}(\hat{y}_o - 4.8). \tag{6}$$

The variance term encourages distinct per-organ scores rather than five copies of the global score, and the two hinge terms keep local scores away from the boundaries of the open $(1,5)$ interval, where the sigmoid has vanishing gradient.

F. Pair Construction and Sampling

The ranking objective draws on a dedicated pair dataset that is rebuilt every epoch. At pair-construction time we apply a holistic-gap consistency filter (hypothesis H2) that discards any pair whose ground-truth ratings differ by less than `MIN_PAIR_RATING_GAP` $= 0.5$, eliminating near-tie pairs whose ordering is dominated by inter-rater noise. We draw three negatives per anchor and combine two samplers: a weighted sampler with square-root smoothing that rebalances anchor rating buckets so that rare extreme faces appear roughly $3.6\times$ more often per epoch, and a hard sampler that biases candidate selection toward distant rating buckets so each anchor is contrasted against more extreme faces. To regularize the geometry, we apply landmark jitter augmentation with $\sigma = 0.003$ after inter-ocular normalization, a perturbation well below the localization error of the landmark detector.

G. Training Configuration

Table I lists the training configuration. The model is implemented in PyTorch with the Deep Graph Library [23] and optimized with Adam [27].

H. Evaluation Protocol

We report global accuracy with the Pearson correlation coefficient (PCC) and mean absolute error (MAE) against the holistic ground-truth score. To quantify fairness, we report the demographic parity difference $\mathrm{DPD} = |\mathrm{MAE}_{\text{Asian}} - \mathrm{MAE}_{\text{Caucasian}}|$. We assess the validity of the local scores with a per-organ Spearman correlation between each predicted local score and its beauty-axis pseudo-score, requiring all five organ correlations to be positive. Finally, because the production model fuses scores by the closed-form softmax of

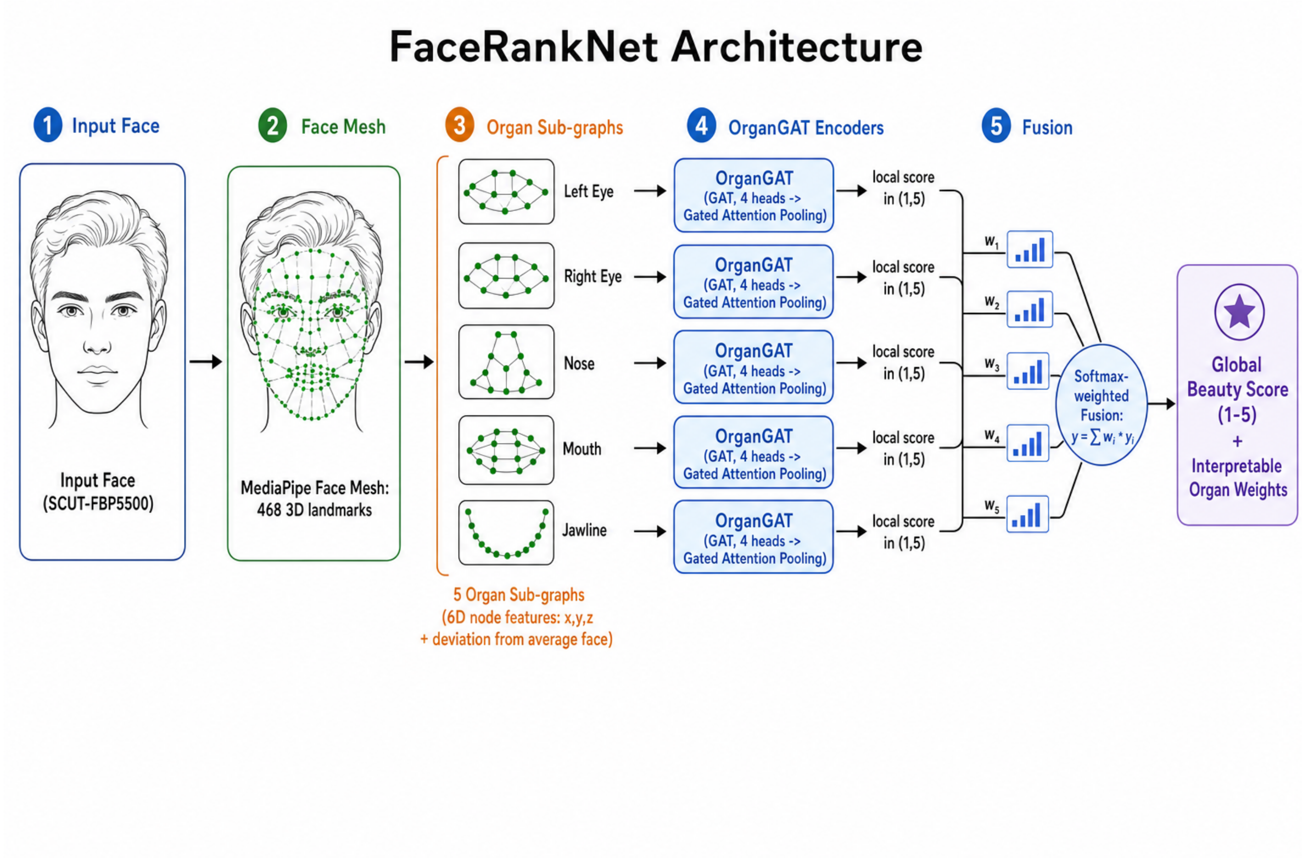

Fig. 2. The FaceRankNet pipeline. A face is reduced to 468 MediaPipe landmarks [8] and partitioned into five organ sub-graphs whose nodes carry six-dimensional features (coordinates plus deviation from the Universal Average Face). Each sub-graph is scored by an independent OrganGAT encoder (graph attention with four heads followed by gated attention pooling), and the five bounded local scores are combined by a learnable softmax into a global beauty score, with the fusion weights serving as an interpretable organ-importance distribution.

TABLE II

TOP-LINE METRICS ON SCUT-FBP5500. THE TEST SET IS EVALUATED ONCE USING THE BEST-VALIDATION CHECKPOINT (EPOCH 49 OF 50).

Split	**PCC** $\uparrow$	**MAE** $\downarrow$	**DPD** $\downarrow$
Validation (best epoch)	0.6745	0.4232	0.0510
Test (1,100 images)	**0.6222**	**0.4410**	**0.0855**

Eq. 2, the organ importance distribution is read directly from $\mathrm{softmax}(w_i)$ rather than estimated by a separate attribution procedure.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Overall Performance

Table II reports the top-line metrics. On the held-out test set, FaceRankNet attains a Pearson correlation of 0.6222, a mean absolute error of 0.4410, and a demographic parity difference of 0.0855. The validation and test gaps are modest, and the training curves in Fig. 3 show that the validation correlation is still rising at the final epoch, indicating that the model is under-fit rather than over-fit and that a longer schedule would likely close part of the remaining gap to higher targets.

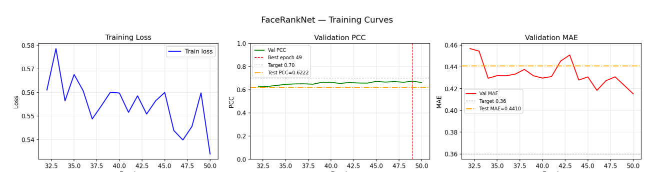

Fig. 3. Training and validation curves. The validation Pearson correlation peaks at epoch 49 and the loss is still drifting downward at epoch 50, indicating that the model is under-trained rather than over-fit.

B. Fairness Across Ethnicity

Table III breaks performance down by ethnicity, and Fig. 4 visualizes the same comparison. The per-ethnicity prototype-and-axis design (H1) keeps the absolute demographic parity difference small at 0.0855. Nonetheless, the Caucasian subgroup is both noisier and less well correlated than the Asian subgroup (PCC 0.454 vs. 0.672; MAE 0.505 vs. 0.419). The smaller Caucasian sample in the test set (278 vs. 822 images) is a contributing factor. We therefore characterize the geometry-only design as one that *structurally reduces* demographic disparity relative to pixel-based predictors, not one that eliminates it.

TABLE III

PER-ETHNICITY BREAKDOWN ON THE TEST SET (HYPOTHESIS H1).

Group	n	PCC ↑	MAE ↓
Asian	822	0.672	0.419
Caucasian	278	0.454	0.505

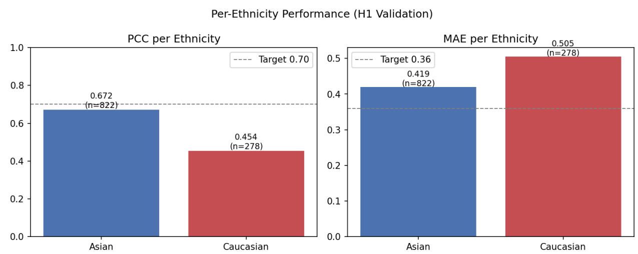

Fig. 4. Per-ethnicity metrics on the test set. The per-ethnicity beauty-axis design keeps the demographic parity difference small in absolute terms, but the Caucasian subgroup remains harder to predict.

C. Pseudo-Label Generator Ablation

The choice of weak-supervision signal is consequential. Table IV compares pseudo-label generators by their global Spearman correlation with human ratings. The directional beauty-axis methods dominate: the quantile-remapped axis reaches $\rho = 0.577$, more than double the best distance-based variant (RMSE to a beauty prototype, $\rho = 0.296$). Strikingly, a pure symmetry heuristic is *negatively* correlated with attractiveness globally ($\rho = -0.160$), challenging the long-standing assumption that symmetry is a primary driver of facial beauty, and an ensemble that blends the axis with symmetry and proportion canons degrades the signal rather than improving it. These results justify the directional, per-ethnicity beauty-axis design of Section III-C.

D. Organ Importance and Local-Score Validity

Because fusion is the closed-form softmax of Eq. 2, the learned weights are an interpretable organ-importance distribution. Fig. 5 shows the per-organ scores for representative faces, demonstrating that the local scores are genuinely distinct rather than degenerate copies of the global prediction. For example, a face with a global score of 2.34 receives organ scores spanning $[1.1, 2.5]$, with the eyes and jawline identified as the weakest components, whereas a face scoring 3.93 globally is composed of organ scores in $[3.6, 5.0]$ with the right eye and nose near the ceiling. This non-degeneracy is the primary qualitative evidence that the part-based design yields faithful, per-organ explanations, and all five organs pass the positive Spearman validity check against their beauty-axis pseudo-scores.

E. Error Structure and Regression to the Mean

Table V reports the mean signed error within each rating bucket, and Figs. 6–7 visualize the same phenomenon. The pattern is a textbook regression to the mean: the lowest bucket is over-predicted by $+0.60$ on average while the highest bucket is under-predicted by -0.82, and the central buckets are well

TABLE IV

PSEUDO-LABEL GENERATORS RANKED BY GLOBAL SPEARMAN ρ WITH HUMAN RATINGS. DIRECTIONAL (AXIS) METHODS DOMINATE DISTANCE- AND RULE-BASED METHODS.

Generator	Category	Global ρ
Axis (quantile remapped)	Directional	$+0.577$
Axis (global ethnicity)	Directional	$+0.574$
Local axis	Directional	$+0.527$
Ensemble (axis + canons + sym.)	Blend	$+0.430$
Canons (proportions)	Rule-based	$+0.350$
RMSE (beauty prototype)	Distance	$+0.296$
RMSE (population mean)	Distance	-0.124
Symmetry	Rule-based	-0.160

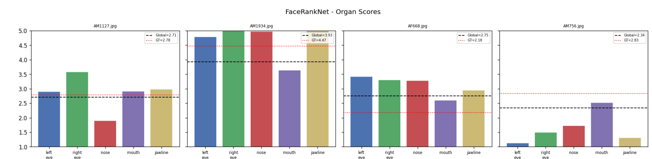

Fig. 5. Per-organ scores for representative test faces. The five local scores are clearly distinct from one another and from the global score, providing interpretable region-level explanations rather than a single collapsed value.

calibrated. The predicted score distribution (Fig. 7) is correspondingly narrower than the ground truth, with essentially no predicted mass above 4.3 even though 126 test faces are rated ≥ 4. The scatter of predictions against ground truth (Fig. 8) shows the same inward compression of the extremes. Because the long tails dominate the correlation, this compression is the main factor capping the achievable PCC, and it is compounded by the under-training visible in Fig. 3.

V. DISCUSSION

The three interpretability methods of Ibrahim *et al.* [1] converge on the eyes and nose as the most influential regions for full-face CNNs, with the chin and lips least important. FaceRankNet offers a complementary, intrinsic view: its learned softmax weights assign the largest aesthetic contribution to the eye and nose sub-graphs, qualitatively consistent with that post-hoc consensus, but obtained without a separate attribution stage and accompanied by per-organ scores for every individual prediction. The agreement is reassuring precisely because the two approaches differ so completely in their inputs and machinery: one aggregates pixel-level saliency from an image CNN, while the other reads convex-combination weights from a geometry-only graph network. Where they diverge, the divergence is informative, since post-hoc saliency can be confounded by texture and context that a landmark model never sees.

It is also worth placing the absolute accuracy in context. FaceRankNet's geometry-only test correlation of 0.6222 is essentially on par with the GNN-only branch of the hybrid GeoFusion-Net (PC $= 0.612$) [17], which likewise reasons purely over facial landmarks. Full hybrid models that add a pixel branch reach correlations near 0.92, but they do so by reintroducing exactly the photometric appearance inputs and

TABLE V
MEAN SIGNED ERROR (PREDICTION MINUS GROUND TRUTH) PER RATING BUCKET ON THE TEST SET.

Bucket	n	Signed error	Reading
Low (1–2)	43	$+0.60$	over-predicts
Below avg. (2–3)	610	$+0.03$	well calibrated
Above avg. (3–4)	321	-0.31	under-predicts
High (4–5)	126	-0.82	strongly under-predicts

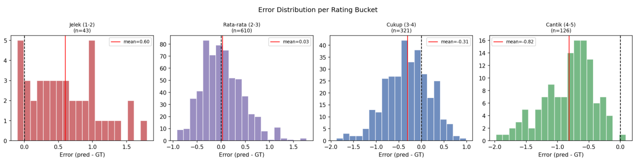

Fig. 6. Signed prediction error by rating bucket. Extreme buckets are pulled toward the center, the signature of regression to the mean.

the attendant opacity and demographic dependence that FaceRankNet deliberately forgoes. We therefore read our results not as a bid for state-of-the-art accuracy but as evidence that a compact, interpretable, geometry-only model captures most of the signal available from facial structure alone.

Two limitations temper the results. First, the geometry-only design reduces but does not remove demographic disparity; the Caucasian subgroup remains harder to predict, partly because of its smaller sample. Second, the regression-to-the-mean behavior, amplified by visible under-training, caps the correlation by compressing the rating extremes. Both are concrete and addressable, and we treat them as the starting point for the agenda below rather than as fundamental barriers. Importantly, the model's intrinsic interpretability makes these failure modes diagnosable at the level of individual organs, which is exactly the property that a responsible deployment of beauty prediction would require.

VI. Future Work and Safety

FaceRankNet is, by design, a step toward beauty prediction that can be inspected and held accountable, and the most valuable directions for future work are those that strengthen both its accuracy and its safety profile.

On the modeling side, the clearest opportunity is to address the regression-to-the-mean compression documented in Table V. Because the validation correlation was still rising at the final epoch, a longer training schedule with learning-rate warmup and decay is the first remedy, followed by loss reweighting or distributional targets that place more emphasis on the sparsely populated extreme buckets. Predicting the full rating distribution rather than its mean, as prior work has shown to help full-face models [4], is a natural extension that would also yield calibrated uncertainty.

On the fairness side, the residual gap between the Asian and Caucasian subgroups must be closed rather than merely reported. This calls for larger and more balanced training data across ethnicities, poses, and expressions; for evaluation on more diverse in-the-wild benchmarks such as MEBeauty [3];

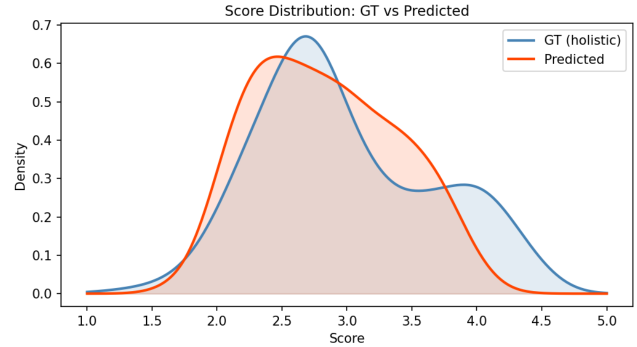

Fig. 7. Kernel density estimates of predicted versus ground-truth scores. The predicted distribution is narrower, with little mass at the high-score tail.

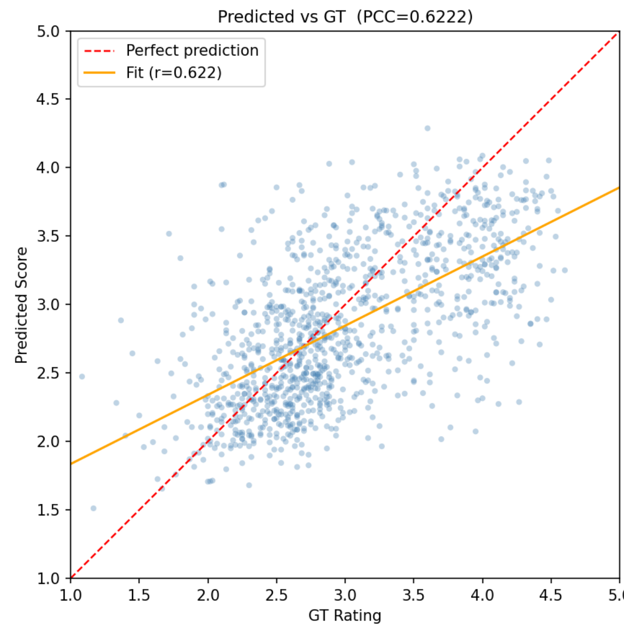

Fig. 8. Predicted versus ground-truth scores on the test set. The inward compression of the extremes limits the attainable correlation.

and for explicit fairness objectives during training [28], [29]. The per-ethnicity beauty-axis mechanism should itself be audited, since fixing an attractiveness direction per group encodes a normative assumption about that group's aesthetics that warrants scrutiny rather than acceptance, especially given evidence that beauty is not perceived uniformly across cultures [13].

The safety considerations are not secondary to the technical ones. Automated facial beauty scoring is a dual-use technology: the same model that could support reconstructive surgery planning could equally be repurposed to rank, filter, or discriminate against people, with documented potential for psychological harm and the reinforcement of narrow, culturally specific beauty norms. We argue for several concrete safeguards. Training data must be sourced with informed consent and clear provenance, and models trained on a given population should not be deployed on populations they were never validated against. Deployed systems should expose calibrated uncertainty and abstain on out-of-distribution or low-confidence inputs rather than emit a spurious score, and high-stakes uses should keep a qualified human in the loop. Scoring or ranking the attractiveness of minors should be categorically out of scope. Finally, the interpretability that motivates FaceRankNet is itself a safety asset: per-organ scores and explicit fusion weights make it possible to audit *why* the model assigned a score, to detect when it is keying on a spurious geometric correlate, and to give a subject a legible account of a decision that affects them [7]. We recommend that any practical FBP deployment pair such intrinsic interpretability with documented intended-use boundaries and ongoing bias monitoring.

VII. Conclusion

We presented FaceRankNet, an interpretable part-based graph attention network that predicts facial beauty from landmark geometry rather than pixels. By scoring five anatomical organ sub-graphs independently and fusing them through a learnable softmax, the model produces a global beauty prediction together with a faithful, per-organ importance distribution as a first-class output. On SCUT-FBP5500 it reaches a test Pearson correlation of 0.6222, a mean absolute error of 0.4410, and a demographic parity difference of 0.0855, and its learned organ weights echo the eyes-and-nose consensus of prior post-hoc interpretability studies while being available by construction. A weak-supervision signal based on per-ethnicity beauty-axis projection proved decisively stronger than distance- and symmetry-based alternatives. The principal limitations, a regression-to-the-mean compression of the rating extremes amplified by under-training, and a residual fairness gap for the smaller Caucasian subgroup, are concrete and addressable. Coupled with the safety agenda we outline, FaceRankNet's intrinsic interpretability offers a foundation for facial beauty prediction that is not only more transparent but also more auditable and more responsibly deployable.

References

[1] A. A. Ibrahim, N. H. Ugail, and H. Ugail, "Is facial beauty in the eyes? a multi-method approach to interpreting facial beauty prediction in machine learning models," *Discover Artificial Intelligence*, vol. 5, no. 16, 2025.

[2] L. Liang, L. Lin, L. Jin, D. Xie, and M. Li, "SCUT-FBP5500: A diverse benchmark dataset for multi-paradigm facial beauty prediction," in *Proc. 24th Int. Conf. Pattern Recognition (ICPR)*. IEEE, 2018, pp. 1598–1603.

[3] I. Lebedeva, Y. Guo, and F. Ying, "MEBeauty: A multi-ethnic facial beauty dataset in-the-wild," *Neural Computing and Applications*, vol. 34, pp. 1–15, 2022.

[4] A. A. Ibrahim, N. H. Ugail, T. H. Jayatileke, M. H. Saffery, and H. Ugail, "Facial beauty analysis using distribution prediction and CNN ensembles," in *Proc. 15th Int. Conf. Software, Knowledge, Information Management and Applications (SKIMA)*. IEEE, 2023, pp. 130–135.

[5] F. Bougourzi, F. Dornaika, and A. Taleb-Ahmed, "Deep learning based face beauty prediction via dynamic robust losses and ensemble regression," *Knowledge-Based Systems*, vol. 242, p. 108246, 2022.

[6] D. E. Boukhari, A. Chemsa, and R. Ajgou, "Facial beauty prediction based on vision transformer," *International Journal of Electrical and Electronic Engineering & Telecommunications*, 2023.

[7] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.

[8] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, "Real-time facial surface geometry from monocular video on mobile GPUs," *arXiv preprint arXiv:1907.06724*, 2019.

[9] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.

[10] G. Rhodes, F. Proffitt, J. M. Grady, and A. Sumich, "Facial symmetry and the perception of beauty," *Psychonomic Bulletin & Review*, vol. 5, pp. 659–669, 1998.

[11] J. H. Langlois and L. A. Roggman, "Attractive faces are only average," *Psychological Science*, vol. 1, no. 2, pp. 115–121, 1990.

[12] A. Sisti, N. Aryan, and P. Sadeghi, "What is beauty?" *Aesthetic Plastic Surgery*, vol. 45, no. 5, pp. 2163–2176, 2021.

[13] J. Zhan, M. Liu, O. G. B. Garrod, C. Daube, R. A. A. Ince, R. E. Jack, and P. G. Schyns, "Modeling individual preferences reveals that face beauty is not universally perceived across cultures," *Current Biology*, vol. 31, no. 10, pp. 2243–2252, 2021.

[14] J. Iyer, R. Nersisson, R. Zhuang, A. N. Joseph Raj, and I. Refayee, "Machine learning-based facial beauty prediction and analysis of frontal facial images using facial landmarks and traditional image descriptors," *Computational Intelligence and Neuroscience*, vol. 2021, p. 4423407, 2021.

[15] T. Peng, M. Li, F. Chen, Y. Xu, and D. Zhang, "Geometric prior guided hybrid deep neural network for facial beauty analysis," *CAAI Transactions on Intelligence Technology*, vol. 9, no. 2, pp. 467–480, 2024.

[16] Y. Zhai, Y. Huang, Y. Xu, J. Gan, H. Cao, W. Deng, R. D. Labati, V. Piuri, and F. Scotti, "Asian female facial beauty prediction using deep neural networks via transfer learning and multi-channel feature fusion," *IEEE Access*, vol. 8, pp. 56 892–56 907, 2020.

[17] D. E. Boukhari and M. Kezai, "GeoFusion-Net: A hybrid graph and convolutional network for facial beauty prediction," in *Proc. IEEE Int. Conf. (in press)*. IEEE, 2025.

[18] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[19] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.

[20] A. Kapishnikov, T. Bolukbasi, F. B. Viégas, and M. Terry, "Segment integrated gradients: Better attributions through regions," *arXiv preprint arXiv:1906.02825*, 2019.

[21] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[22] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?" explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.

[23] M. Wang, D. Zheng, Z. Ye, Q. Gan, M. Li, X. Song, J. Zhou, C. Ma, L. Yu, Y. Gai, T. Xiao, T. He, G. Karypis, J. Li, and Z. Zhang, "Deep graph library: A graph-centric, highly-performant package for graph neural networks," *arXiv preprint arXiv:1909.01315*, 2019.

[24] Y. Li, D. Tarlow, M. Brockschmidt, and R. Zemel, "Gated graph sequence neural networks," *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.

[25] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 794–803.

[26] C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender, "Learning to rank using gradient descent," in *Proc. 22nd Int. Conf. Machine Learning (ICML)*, 2005, pp. 89–96.

[27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.

[28] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," *Proc. 3rd Innovations in Theoretical Computer Science Conf. (ITCS)*, pp. 214–226, 2012.

[29] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Computing Surveys*, vol. 54, no. 6, pp. 1–35, 2021.